

Benjamin Klugah-Brown, Ph.D.

Distinguished Associate Professor of Biomedical Engineering

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PROFILE

Neuroimaging research scientist with a multidisciplinary foundation in biomedical engineering and computer science. My work integrates advanced neuroimaging, computational modeling, and meta-analytic approaches to investigate the neural mechanisms of psychiatric and neurological disorders, including addiction, reward-based dysfunction, depression, and other affective conditions. I apply data-driven methods, artificial intelligence, machine learning, and graph-theoretical analysis to large-scale resting-state and task-based fMRI datasets and develop translational software tools for mental health assessment and intervention. Author and co-author of more than 80 peer-reviewed publications in high-impact journals.

CURRENT APPOINTMENTS

Distinguished Associate Professor (Research) — China–Cuba Belt and Road Joint Laboratory on Neurotechnology and Brain-Apparatus Communication-UESTC

Group Leader — BrainInformatics and Reward Systems (BIRs), Brain Connectivity Lab

Joint Appointment — School of Life Science and Technology, UESTC

Core Member — MIND & AI Lab (China / Hong Kong)

Associate Editor — Frontiers in Neuroimaging

EDUCATION

Ph.D., Biomedical Engineering 2018
University of Electronic Science and Technology of China (UESTC) — *Computational Neuroscience, with application to frontal lobe epilepsy*

M.S., Computer Science and Engineering 2015
University of Electronic Science and Technology of China (UESTC) — *Algorithm Formulation*

B.S., Computer Science 2010
University for Development Studies — *Software Design and Implementation*

RESEARCH INTERESTS

- Neural mechanisms of addiction, reward dysfunction, and affective disorders (depression and related conditions)
- Coordinate-based and image-based meta-analysis of functional and structural neuroimaging
- Resting-state and task-based fMRI: connectivity mapping and co-activation pattern analysis
- Graph-theoretical analysis and brain network hub characterization
- Artificial intelligence and machine learning applied to large-scale neuroimaging datasets
- Translational software tools

GRANTS & FUNDING

Research Fund for International Young Scientists (RFIS) 2023–2024
National Natural Science Foundation of China (NSFC) — Research Fund for International Scientists · Principal Investigator

Project: Investigating the structural and functional MRI characteristics of internet gaming disorder in young adults via a naturalistic experimental paradigm

EDITORIAL & PROFESSIONAL SERVICE

- **Associate Editor**, *Frontiers in Neuroimaging*
- Peer reviewer for international journals in neuroimaging, psychiatry, and computational neuroscience

TEACHING & SUPERVISION

Supervises doctoral and master's students in biomedical engineering, with specialized graduate teaching in the fundamentals of brain science and brain imaging.

REPRESENTATIVE PUBLICATIONS

Author name in bold; (CA) denotes corresponding author.

Peer-Reviewed Journal Articles

1. Wu, Hongzhou; Xiao, Jinming; Agoalikum, Elijah; Baig, Talha Imtiaz; Becker, Benjamin; Ferraro, Stefania; Biswal, Bharat B; **Klugah-Brown, Benjamin** ^(CA); Maes, Michael (2026). Multi-Indicator Entropy Hub Score: A quantitative approach to hub analysis in brain networks. *NeuroImage*, 121799.
2. **Klugah-Brown, Benjamin**; Li, Yuan; Wu, Hao; Yang, Zhenzhen; Hagan, Abraham Tonny; Agoalikum, Elijah; Wang, Pan; Biswal, Bharat B (2026). Dynamic brain connectivity patterns induced by oxytocin: An fMRI co-activation pattern analysis study. *Molecular Psychiatry*, 1–13.
3. Yang, Zhenzhen; **Klugah-Brown, Benjamin** ^(CA); Ding, Guobin; Zhou, Wenchao; Biswal, Bharat B (2025). Brain structural differences in cocaine use disorder: Insights from multivariate and neurotransmitter analyses. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 136, 111159.
4. Wu, Hongzhou; Yang, Zhenzhen; Cao, Qingquan; Wang, Pan; Biswal, Bharat B; **Klugah-Brown, Benjamin** ^(CA) (2024). MQGA: A quantitative analysis of brain network hubs using multi-graph theoretical indices. *NeuroImage*, 303, 120913.
5. **Klugah-Brown, Benjamin**; Wang, Pan; Jiang, Yuan; Becker, Benjamin; Hu, Peng; Uddin, Lucina Q; Biswal, Bharat (2023). Structural–functional connectivity mapping of the insular cortex: A combined data-driven and meta-analytic topic mapping. *Cerebral Cortex*, 33(5), 1726–1738.
6. **Klugah-Brown, Benjamin**; Yu, Yue; Hu, Peng; Agoalikum, Elijah; Liu, Congcong; Liu, Xiqin; Yang, Xi; Zeng, Yixu; Zhou, Xinqi; Yu, Xin (2022). Effect of surgical mask on fMRI signals during task and rest. *Communications Biology*, 5(1), 1004.
7. **Klugah-Brown, Benjamin**; Zhou, Xinqi; Wang, Lan; Gan, Xianyang; Zhang, Ran; Liu, Xiqin; Song, Xinwei; Zhao, Weihua; Biswal, Bharat B; Yu, Fangwen (2022). Associations between levels of Internet Gaming Disorder symptoms and striatal morphology: Replication and associations with social anxiety. *Psychoradiology*, 2(4), 207–215.
8. **Klugah-Brown, B**; Zhou, X; Pradhan, BK (2022). Erratum: Common neurofunctional dysregulations characterize obsessive–compulsive, substance use, and gaming disorders — An activation likelihood meta-analysis of functional imaging studies (vol 26, e12997, 2021). *Addiction Biology*, 27(6). [Erratum]
9. **Klugah-Brown, Benjamin**; Jiang, Chenyang; Agoalikum, Elijah; Zhou, Xinqi; Zou, Liye; Yu, Qian; Becker, Benjamin; Biswal, Bharat (2021). Common abnormality of gray matter integrity in substance use disorder and obsessive–compulsive disorder: A comparative voxel-based meta-analysis. *Human Brain Mapping*, 42(12), 3871–3886.
10. **Klugah-Brown, Benjamin**; Zhou, Xinqi; Pradhan, Basant K; Zweerings, Jana; Mathiak, Klaus; Biswal, Bharat; Becker, Benjamin (2021). Common neurofunctional dysregulations characterize obsessive–compulsive, substance use, and gaming disorders: An activation likelihood meta-analysis of functional imaging studies. *Addiction Biology*, e12997.

11. **Klugah-Brown, Benjamin;** Di, Xin; Zweerings, Jana; Mathiak, Klaus; Becker, Benjamin; Biswal, Bharat (2020). Common and separable neural alterations in substance use disorders: A coordinate-based meta-analysis of functional neuroimaging studies in humans. *Human Brain Mapping*.